Fleet Data Analysis and Compliance Report

1. Summary

This project analyzes fleet data for 20 employees across four departments in an organization. The objective was to evaluate gas usage against allocated allowances, review vehicle compliance, analyze service costs and mileage and identify areas that require management attention.

I used Excel formulas to calculate remaining gas allowances, classify usage, estimate costs, assess compliance and flag records requiring follow-up. I also used PivotTables to summarise gas usage, service costs and kilometres travelled by department.

The results identified five employees who exceeded their individual gas allowances and seven employees whose vehicle records did not meet the defined compliance requirements. Sales recorded the highest service costs, while Operations recorded the highest total gas usage.

2. Project Objectives

The analysis aimed to:

- Compare individual gas usage against allocated allowances.
- Identify employees exceeding their gas allowances.
- Calculate and compare vehicle service costs across departments.
- Review vehicle documentation and compliance status.
- Identify records requiring follow-up.
- Summarise departmental gas usage, service costs and mileage using PivotTables.

3. Methodology

The dataset contained 20 employee records and information about departments, vehicles, gas allowances, gas usage, monthly mileage, service costs, licences, insurance, vehicle documentation and CRM status.

I used Excel formulas to automate calculations and classify records. PivotTables were then used to summarise the results by department.

The main calculations included:

- **Remaining gas allowance:** Gas allowance minus gas used.
- **Gas usage status:** Whether usage exceeded the individual allowance.
- **Compliance status:** Whether the licence and insurance were valid and vehicle documentation was complete.
- **Action required:** Whether a record needed follow-up based on its compliance status.

4. Gas Usage Analysis

The combined gas allowance across all employees was 31,100 units, while recorded gas usage totalled 29,070 units. This left a combined balance of 2,030 units.

Gas Usage Measure	Result
Total Allowance Allocate	31,100 units
Total Gas Used	29,070 units
Remaining Allowance	2030 units
Employees Within Allowance	15
Employees Over Allowance	5

Although total usage was below the combined allowance, five employees exceeded their individual allocations. This demonstrates why fleet usage should be monitored at both the overall and individual levels.

Recommendation: Review the five over-allowance records to establish whether the exceptions were caused by higher mileage, operational requirements, unusual consumption or inaccurate records.

5. Departmental Analysis

The PivotTable summarised service costs, gas usage and monthly kilometres by department.

Department	Service Costs	Gas Used	Monthly Kilometers
Sales	R16,100	R12,675	R26,460
Operations	R3,500	R8,565	R18,370
Finance	R5,000	R4,290	R9,615
HR	R3,500	R3,540	R8,180
Total	R28,100	R29,070	R62,625

Key findings

- **Sales** recorded the highest service costs at R16,100, approximately 57.3% of total recorded service costs.
- **Sales** also recorded the highest gas usage and monthly mileage.
- **Operations** recorded the second-highest gas usage and mileage, despite having R3,500 in service costs.
- **HR** recorded the lowest gas usage and monthly mileage.

The higher usage and mileage in Sales may be related to its operational travel requirements. However, the dataset alone does not establish the cause. Further analysis would be needed to compare costs and usage fairly across vehicles and departments.

Recommendation: Review Sales and Operations vehicle usage, service histories and mileage patterns to identify opportunities for better planning and cost control.

6. Vehicle Compliance Analysis

Compliance was assessed using three conditions: a valid license, valid insurance and complete vehicle documentation.

Compliance Measure	Employees
Compliant	13
Non-Compliant	7
Total	20

Seven employees were flagged for follow-up because at least one of the required compliance conditions was not met.

The records included expired licenses, expired insurance and missing vehicle documentation. Missing documentation was particularly common, appearing in seven employee records.

Recommendation: Prioritize the seven flagged records, verify outstanding documents and update the tracking system once issues have been resolved. Any expired license or insurance should be reviewed promptly in accordance with applicable requirements and company policy.

7. Limitations and Data Quality Considerations

The findings should be interpreted in light of several limitations:

- The dataset contains only 20 employee records, so it may not represent the entire fleet.
- The gas usage unit and its relationship to monetary fuel expenditure need to be confirmed.
- The current total monthly cost formula adds gas usage to service cost. These values should only be combined if they represent compatible monetary amounts.
- Cost per kilometre should be recalculated after confirming the fuel-cost data and other relevant operating costs.
- Departmental differences may reflect different travel requirements rather than inefficiency.
- The analysis identifies compliance exceptions but does not establish whether they have already been resolved outside the dataset.

8. Conclusion

The analysis demonstrates how Excel can be used to turn raw fleet records into actionable operational insights. The workbook identified individual gas allowance exceptions, highlighted compliance issues and provided a departmental overview of service costs, gas usage and mileage.

The main priorities are to investigate individual allowance exceptions, resolve outstanding compliance issues and review the high service costs recorded by Sales. Confirming the units and financial calculations would further strengthen the reliability of the analysis.